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Optimized Stock Price Forecasting: A Deep Learning Approach Using QCTO Based GRU Network

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***Abstract*—Stock represents ownership in a publicly traded company and stock price is the current market value of a company's stock. It fluctuates based on factors such as supply and demand, company performance, and overall market conditions. Predicting accurate stock price is the basic task of every trader and investor to gain profit from the financial market. Traditional trend prediction models are generally based on experts knowledge and experience. Often, it is difficult to find out hidden patterns contained in the stock time series data due to its inherent volatility and complexity. To address this issue, a Quantum Class Topper Optimization based Gated Recurrent Unit (QCTO-GRU) model is proposed to predict stock price movement, highlighting its ability to capture intricate patterns in dynamic market conditions. The parameters of GRU model are optimized using QCTO, a meta-heuristic optimization technique. The proposed model excels in predicting the stock price and outperforms the existing models with the value of 0.01271 and 0.00975 for MAE and MSE respectively.**

***Index Terms*—Stock Price Prediction, Gated Recurrent Unit, Quantum Computing, Class Topper Optimization.**

I. Introduction

Stock market prediction has long been a critical research area in financial markets, playing a significant role in guiding investment strategies for traders, analysts, and institutional investors. Accurately forecasting stock prices can lead to substantial financial gains, yet it remains one of the most challenging problems in economics due to the inherent volatility, nonlinearity, and complex dependencies in financial data. Traditional time-series forecasting methods such as Autoregressive Integrated Moving Average (ARIMA) and linear regression [14] have been widely used in past decades, but they often struggle to capture the intricate patterns and rapid fluctuations in stock price movements. With the rapid advancement of artificial intelligence and Deep Learning (DL), modern Machine Learning (ML) [16] techniques have emerged as powerful tools for improving the accuracy of stock price prediction.

For financial time-series forecasting, DL models—in particular, Recurrent Neural Networks (RNNs) and their variations—have drawn a lot of interest. Long Short-Term Memory (LSTM) [15] networks and Gated Recurrent Units (GRU) have demonstrated strong capabilities in capturing long-term dependencies in sequential data, overcoming the vanishing gradient problem that limits traditional RNNs. Additionally, bidirectional architectures such as BiLSTM and BiRNN have been utilized [10] to enhance predictive performance by learning from both past and future price trends.

Beyond RNN-based architectures, hybrid models that integrate multiple DL techniques have also demonstrated impressive results. CNN-BiLSTM-Attention [9] models have been proposed to leverage Convolutional Neural Networks (CNNs) for feature extraction, BiLSTM for sequential pattern recognition, and attention mechanisms for dynamically weighting relevant information. Some studies have explored sentiment-enhanced predictive models. For instance, the SA-DLSTM model [5] integrates an Emotion-Enhanced Convolutional Neural Network (ECNN) to extract sentiment representation from user-generated comments.

Despite these advancements, stock market forecasting remains a highly challenging task due to external uncertainties such as economic shifts, geopolitical events, and sudden market sentiment changes. The Main contributions of the proposed work can be found as follows.

A DL based approach for stock price prediction is presented based on the GRU network integrating with Quantum CTO. The time series problem in regression is solved using Quantum computing theory. Also, a highly efficient and scalable data preprocessing technique is accumulated to explore intricate patterns in the data. Various evaluation metrics are employed to evaluate the performance of proposed QCTO-GRU model.

The rest of the paper is organized as follows: Section II presents the Literature Review. Section III describes the Proposed Methodology. Section IV discusses the Experimental Results. Finally, Section V concludes the paper and outlines future research directions.

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II. Literaure Review

The study in [1] compared six DL models—DNN, RNN, LSTM, Bi-LSTM, GRU, and CNN—for long-term daily stock price forecasting of five global indices (Nifty, DJIA, DAX, Nikkei 225, and SSE). Results showed that LSTM performed best overall, while CNN overfitted the training data and failed in testing. The study highlighted the challenge of predicting long-term stock prices and emphasized LSTM's ability to capture temporal dependencies, making it a valuable tool for investors. However, the model may miss rapid market fluctuations due to dependent on long-terms trends.

The authors [2] introduced the VIX-Lasso-GRU model for predicting intraday stock index futures prices in China. By incorporating the volatility index (VIX) for market sentiment and using the Lasso algorithm for feature selection, the model improved both accuracy and training speed compared to pure GRU and LSTM models. Tested on 5-minute closing prices of three major index futures (SSE 50, CSI 300, CSI 500), the model outperformed traditional approaches, demonstrating that implied volatility and dimensionality reduction play crucial roles in high-frequency financial forecasting.

The HFSLSMR-LSTM model proposed by [3] integrated hierarchical feature selection with local shuffling (HFSLS) and model reweighting (MR) for stock price forecasting. The method improved feature selection by reducing redundancy and enhancing key feature retention at each layer. Evaluated on stock index and Dow Jones Industrial Average datasets, it outperformed Informer, DoubleEnsemble, LSTM, GRU, BI-GRU, and BI-LSTM in terms of various regression evaluation metrics. The outcome demonstrated the model's effectiveness in handling multi-featured, high-volatility financial data effectively.

Stock price forecasting using DL and text analysis was explored by [6], focusing on the halal tourism sector. A World Halal Tourism Composite Sentiment Index (WHTCSI) was developed to integrate sentiment analysis into predictions. Among the tested models, CNN outperformed others, demonstrating strong forecasting ability even at the country level. Explainable AI analysis revealed that the sentiment index contributed 35.55% to the predictions. The findings provided valuable insights for investors and analysts in decision-making. However, the performance heavily relies on the accuracy of social media sentiment interpretation.

The authors [7] addressed the interpretability challenges in stock price prediction by proposing multi stage approach to select features that captures both the collective behavior of features over time and their time-dependent nature. It introduced best feature and piecewise best feature subset selection for long-term and short-term dynamics respectively. Using hybridized genetic algorithm-ML regressions, the method improved forecasting accuracy, reducing RMSE for optimal features and internal technical indicators by 10.42% and 13.47% respectively. Though this method may ignore the sequential nature of financial time series.

III. Proposed Methodology

A. Model baseline: GRU

In solving the temporal problem or vanishing gradient problem in native neural network model, RNN is one of the most widely adopted models. It can process lengthy sequences of data like Stock prices. LSTM and GRU are the type of RNN, which can handle historical data to get intricate patterns using different gate structures. GRUs [2] are an advanced type of RNNs designed to handle sequential data efficiently. Unlike traditional RNNs, GRU uses two key gates— update and reset, to control the flow of information. The update gate helps retain important past information, ensuring that relevant trends from previous time steps are carried forward [10]. Meanwhile, the reset gate determines how much past data should be ignored, enabling the model to concentrate on the most essential trends. This structure makes GRUs simpler and computationally more efficient than LSTMs while maintaining strong predictive performance, making them well-suitable for stock price forecasting. Architecture of the proposed QCTO-GRU model shown in Fig. 1.

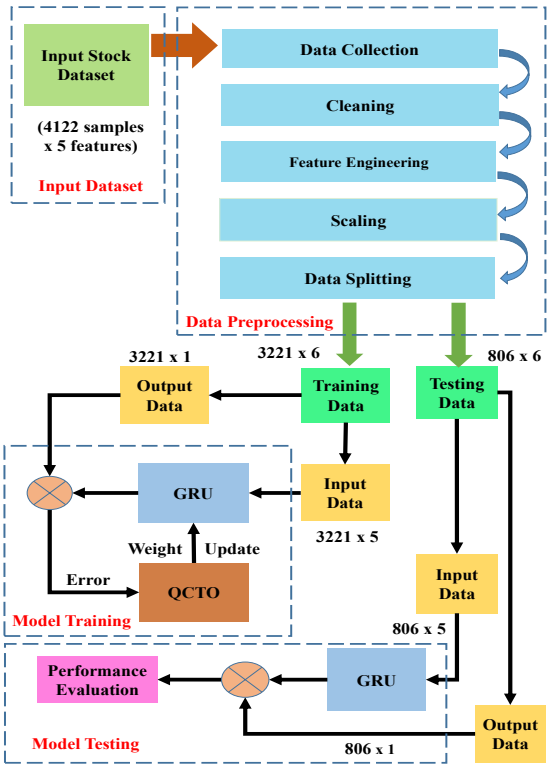

Fig. 1. Architecture of QCTO-GRU Model

In financial market prediction, GRUs have proven to be highly effective in capturing complex stock price movements. Compared to traditional ML models like Random

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Forest and ARIMA, GRUs handle long-term dependencies much better without excessive memory usage [17].

The output and new cell state are determined using the following expressions.

$$\mathcal{R}_t = \sigma_g \left(\mathcal{X}_t \cdot \omega_{xr} + \mathcal{H}_{t-1} \cdot \omega_{hr} + b_r \right), \tag{1}$$

$$\mathcal{U}_t = \sigma_g \left(\mathcal{X}_t \cdot \omega_{xu} + \mathcal{H}_{t-1} \cdot \omega_{hu} + b_u \right), \tag{2}$$

$$\mathcal{C}_S = tanh \left(\mathcal{X}_t \cdot \omega_{xc} + (\mathcal{R}_t \odot \mathcal{H}_{t-1}) \cdot \omega_{hc} + b_h \right), \tag{3}$$

$$\mathcal{H}_t = \mathcal{U}_t \odot \mathcal{H}_{t-1} + (1 - \mathcal{U}_t) \odot \mathcal{C}_S \,. \tag{4}$$

Where $\mathcal{R}_t$ is the reset gate, $\mathcal{U}_t$ is the update gate, $\mathcal{C}_S$ is intermediate cell state, $\odot$ is the element wise multiplication and σ_g is sigmoid function. $\mathcal{X}_t$ is the input data, $\mathcal{H}_{t-1}$ is the hidden state, ω_{xr}, ω_{hr}, ω_{xu}, ω_{hu}, ω_{xc}, ω_{hc} are the weights of the model and b_r, b_u, b_c are the biases.

B. QCTO

QCTO (Quantum Class Topper Optimization) [12] is an advanced meta-heuristic optimization algorithm inspired by how students learn and compete in a classroom. It is an improved version of the CTO algorithm [4], which models knowledge transfer among students, where individuals learn from the best-performing students in each section.

QCTO incorporates quantum computing principles to enhance its optimization capabilities. Concepts such as quantum superposition and entanglement enable better exploration of the solution space [18], addressing challenges that classical methods struggle with. With the rising interest in quantum-based optimization, QCTO represents a significant advancement by integrating quantum computing with an intelligent learning-based approach. This combination improves the algorithm's search efficiency and convergence, making it a powerful tool for solving complex optimization problems.

1) Algorithm:

- **Declaration** Declare the parameters like no. of sections (SE), students (S) in each section, courses (CS) per student, examinations (E) and other constraints.
- **Random Generation of population** In this step, the phase of Q-bit (θ) are randomly generated based on students, sections and courses, where θ ranges between $0° \leq \theta \leq 360°$ and also satisfy following equation.

$$\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right), \tag{5}$$

  whereas, these α and β are the two bits of the quantum bit. In quantum computing, α and β are complex probability amplitudes that define the quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, satisfying eq (6).

$$\alpha^2 + \beta^2 = 1. \tag{6}$$

- **Parameter encoding** To get the actual parameter (Q), α and β are encoded using eq (7). Q falls within the range [h, l].

$$Q_{(x,y,z)} = \begin{cases} \alpha^2 * (h - l) + l, & \gamma \leq p, \\ \beta^2 * (h - l) + l, & \gamma > p, \end{cases} \tag{7}$$

  where γ is the random number ranges between [0,1]; p is the state probability of the Q-bits; x is the no. of sections; y is the no. of students; z is the no. of courses.
- **Evaluation** The Q values are processed using a Gated Recurrent Unit (GRU) model to extract meaningful patterns and trends. The output obtained from this process corresponds to the Performance Factor (PF) for each student, as defined by eq (4). This PF value reflects the predicted performance of the students based on the learned representations from the GRU model. Based on these PFs, section best (SB) and the class best (CB) are found. The evaluation procedure is the same as for a basic CTO [4].
- **Phase Updation** In the updation stage, SBs update their learning from the CB using the eq (8).

$$\Delta Q^{t+1} = w^t.\Delta Q^t + C.r1 + (Q^{CB} - Q^{SB}) \tag{8}$$

  And the normal students update their learning from their SB with eq (9).

$$\Delta Q^{t+1} = w^t.\Delta Q^t + C.r2 + (Q^{SB} - Q^{S}) \tag{9}$$

  where ΔQ^t is the phase updation in current examination; ΔQ^{t+1} is the phase updation for next examination; C is the acceleration coefficient; $r1$ and $r2$ are random values ranges between 0 and 1; w^t Q^{CB}, Q^{SB}, and Q^S represents the phases of CB, SB, and S respectively; Improvement factor is updated using:

$$w^t = w_{max} - \left(\frac{w_{max} - w_{min}}{E}\right) \times t, \tag{10}$$

  where w^t is the improvement factor for current examination t; w_{max} and w_{min} are the maximum and minimum weight values respectively; and E is the total number of examinations (iteration).
- **Q-bit updation** The Q-bit is updated using a quantum rotation gate [13], which is a reversible operation with a unit magnitude. This updation is done on the phase correction ΔQ^{t+1} and the previous quantum Q-bits, α and β. The criterion for qubit updation using the quantum rotation gate is given in eq (11).

$$\begin{bmatrix} \alpha^{t+1} \\ \beta^{t+1} \end{bmatrix} = \begin{bmatrix} cos(\Delta Q^{t+1}) & -sin(\Delta Q^{t+1}) \\ sin(\Delta Q^{t+1}) & cos(\Delta Q^{t+1}) \end{bmatrix} \begin{bmatrix} \alpha^t \\ \beta^t \end{bmatrix} \tag{11}$$

- **Parameter decoding** The actual parameters is transformed back into Q-bits using the eq (12).

$$Q^{t+1} = \begin{cases} Q^t * \beta^2, & \gamma \geq p \\ Q^t * \alpha^2, & \gamma < p \end{cases} \tag{12}$$

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TABLE I
DATA DESCRIPTION

Collected dataset	4122 samples x 5 features
Features used	OpenPrice, ClosePrice, LowPrice, HighPrice, and Volume
Preprocessed dataset	4027 x 6
Train data (80%)	3221 x 6
Test data (20%)	806 x 6
Input train data, Output train data	3221 x 5, 3221 x 1
Input test data, Output test data	806 x 5, 806 x 1

where Q^t and Q^{t+1} represents current and next phase parameters respectively.

- **Termination** The process of optimization is terminated when convergence in CB value is achieved or maximum no. of examination is reached else goto step 4.

C. Data collection & preprocessing

In supervised learning, getting proper data and then making it feasible for the model is the most crucial step. This stage is categorized in following 6 steps.

1) Data collection: For experimental purpose, stock price data of Bank Of America is gathered from (https://finance.yahoo.com) website during from January 2005 to October 2021.

2) Cleaning: For many irrational behaviour in the market, data may have some noise which may fail to track the trend. Therefore cleaning of data is necessary. Here, missing data problem is not severe, because trading market is more reliable and well maintained. Forward fill method is used to fill the missing data, and duplicate values also are removed from the dataset.

3) Feature Engineering: Feature Engineering is an important stage where features are first extracted from the cleaned dataframe. The selected features for the proposed model to predict the stock price are represented in Table I.

4) Scaling: In the feature scaling stage, features are scaled within the range 0 to 1 using Min-Max Scaler, ensuring all features contribute equally to the model.

5) Splitting: The entire dataset is bifurcated into 80% and 20% dedicated for train and test datasets respectively.

IV. EXPERIMENTAL RESULTS

The performance of the proposed QCTO-GRU model is evaluated for Bank Of America stock price prediction. The results are presented through convergence behavior, training performance, comparative evaluation, and prediction visualization. Table II presents the optimized hyperparameter values used in the proposed model. The key parameters were fine-tuned using the QCTO algorithm to improve the GRU model's capacity for prediction.

TABLE II
PARAMETERS OF CTO MODEL

Parameters	Values
No. of Section	5
No. of Student	100
No. of Course	3 $\times$ 6 matrix
No. of Examinations	100
w_{max}, w_{min} (weight factor)	0.5, 0
C (acceleration coefficient)	1.2
ISS (improvement factor for ST)	0.001
IST (improvement factor for student)	0.001

The model training is shown using the convergence graphs of the proposed QCTO-GRU model for three performance metrics: MAE, MSE, and MAPE. Figure 2, 3, and 4 illustrate the convergence curves of these metrics over training epochs.

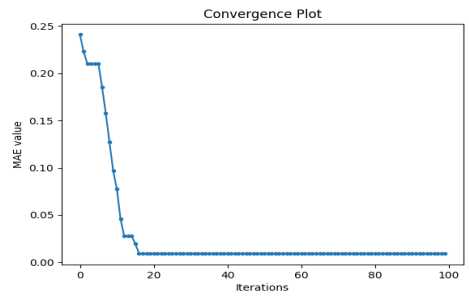

Fig. 2. Convergence Graph for MAE

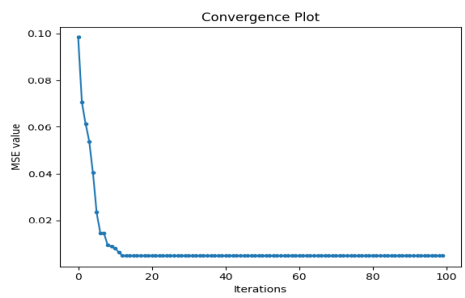

Fig. 3. Convergence Graph for MSE

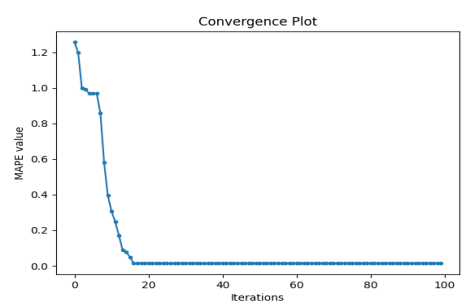

Fig. 4. Convergence Graph for MAPE

The results show, how error values were decline steadily, that indicates the stability and efficiency of the proposed model in optimizing the model parameters.

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TABLE III
TEST RESULTS

Model	MSE	MAE	MAPE
CNN-BiLSTM [9]	0.0128641	0.0177637	0.0198415
Proposed QCTO-GRU	0.0097568	0.0127147	0.0249261

The predictive performance of proposed QCTO-GRU model is compared with another benchmark model CNN-BiLSTM-Attention [9] to validate its effectiveness. Table III demonstrates a comparative analysis based on test results.

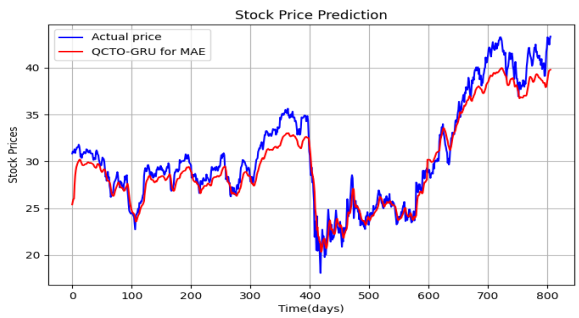

Fig. 5. Prediction Plot for MAE

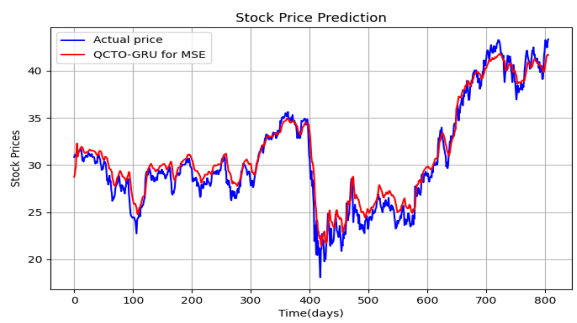

Fig. 6. Prediction Plot for MSE

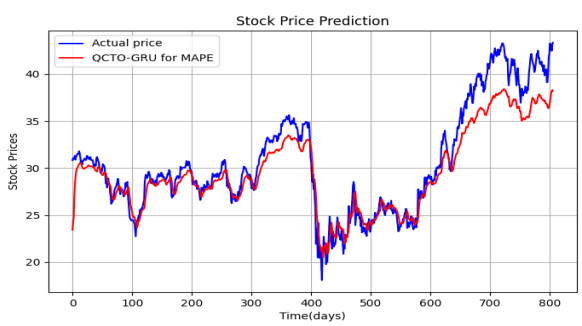

Fig. 7. Prediction Plot for MAPE

The proposed QCTO-GRU model outperforms the baseline model across MAE and MSE metrics, indicating better accuracy, but has slightly higher MAPE, suggesting less consistent percentage error. To further validate the accuracy of the proposed model, three stock price prediction curves are presented, each comparing the predicted values with actual stock prices. Figures 5, 6, and 7 illustrate the high correlation between predicted and actual prices, indicating the robustness of the proposed QCTO-GRU model.

V. CONCLUSION

QCTO-GRU model is proposed for stock price prediction, leveraging Quantum Class Topper Optimization (QCTO) to optimize the GRU network. The results demonstrate that the proposed approach effectively reduces prediction errors, as shown by the low MAE, MSE, and MAPE values. The convergence analysis further confirmed the model's stability and efficiency in learning stock price patterns. The comparative evaluations highlight the superiority of QCTO-GRU over other benchmark model, achieving higher predictive accuracy. Future research could explore the integration of additional financial indicators and alternative DL architectures to further enhance forecasting performance.

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